Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/28/2025
PI Name:	Britten, Richard Ph.D.		
Project Title:	Hadron-induced Impairment of Executive Function: Role of Perturbed Neurotransmission and the Exacerbating Impact of Sleep Deprivation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Brittera@evms.edu	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	757-446-5038
Organization Name:	Eastern Virginia Medical School		
PI Address 1:	Radiation Oncology		
PI Address 2:	700 W Olney Rd		
PI Web Page:			
City:	Norfolk	**State:**	VA
Zip Code:	23507-1607	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-NSCOR Radiation
Start Date:	12/02/2015	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per NASA JSC Grants Office (Ed., 11/27/23). NOTE: End date changed to 12/31/2023 per NSSC info (Ed., 6/9/22). NOTE: End date changed to 5/01/2022 per NSSC info (Ed., 11/30/20) NOTE: End date changed to 12/31/2020 per NSSC info (Ed., 11/12/19) Ed. NOTE (April 2016): Proposal modified from original NSCOR proposal, per Space Radiation Element; not an NSCOR project		
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI) report, Co-Investigator Ashley Blackwell is no longer with the project (Ed., 4/16/25).		
COI Name (Institution):	Sanford, Larry Ph.D. (Eastern Virginia Medical School) Wellman, Laurie Ph.D. (Eastern Virginia Medical School) Yoon, Hargsoon Ph.D. (Norfolk State University)		
Grant/Contract No.:	NNX16AC40G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The proposed studies will address multiple issues of concern to NASA. The proposed studies will generate data on the likelihood that GCR (galactic cosmic radiation) exposure will result in the impairment of neurocognitive (Executive Function) tasks that will be absolutely vital for the successful completion of a deep-space mission, under conditions that are more representative of the actual mission (when individuals are suffering from perturbed sleep). These studies will be conducted in a rat model that is relatively unique in that the rats are exercised regularly, and are preselected for a high level of executive function performance (high cognitive reserved) prior to space radiation exposure. Such a model system more closely resembles the physical and cognitive reserve of astronauts, and addresses a key operational issue of whether space radiation exposure will impact previously imprinted cognitive skills. Specifically, these studies will determine the impact that mission-relevant doses of space radiation (Z<15 single ion exposures, in addition to the multi-ion GCRSim) have on advanced executive functions, specifically Attentional Set Shifting (ATSET) and creative problem solving, or unconstrained cognitive flexibility (UCFlex). This data can be used to address Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders Bmed Gap 102 [previously central nervous system (CNS) Gap 2], and determine the likelihood of astronauts being able to successfully conduct neurocognitive (problem-solving) tasks. This study will also determine whether the incidence and/or severity of executive function impairments would be altered by exposure to another frequent flight stressor, insufficient sleep [Bmed-107 and Bmed-108, formerly CNS Gap 8]. A key component of these studies is an assessment of inter-individual susceptibility to develop deficits in these two cognitive processes following space radiation exposure (with or without the additional sleep stressor). Such data can be used (by others) to determine the best approach to develop a population Threshold Dose for dose risk estimations [Bmed-102 and Bmed-107, formerly CNS Gap 3]. [Ed. note October 2020: Human Research Program (HRP) gaps have changed with Integrated Research Plan Rev L such that gaps listed previously have been merged into newer gaps, as noted] Initially, three hypotheses were tested by the studies covered by this grant. 1. Sleep perturbation reduces the ability of rats to perform executive functions, and will be especially pronounced in space radiation (SR)-irradiated rats. 2. Sleep perturbation alters the sensitivity of the brain to develop SR-induced cognitive impairment. 3. SR irradiation alters Delta and Theta wave activity (EEG), and that these SR-induced changes in EEG activity are the underlying cause of the impairment of executive function performance. In 2019, we received supplemental funding (an HRP-IWS Graduate Fellowship) to develop the technical capability to simultaneously assess neural activity in multiple brain regions (neural network activity) of rats that are actively engaged in cognitive flexibility tasks, and how sleep fragmentation impacts neural network cohesiveness. In 2020, we received further supplemental funding (an HRP-IWS Graduate Fellowship) to establish whether sleep fragmentation exacerbates SR-induced deficits in fine motor skills.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The newly developed capability of measuring neural network functionality when rodents are under cognitive loading will allow more detailed work on establishing the basis for chemo- and beamo-brain in cancer patients, and to help understand how task switching is impaired in age-related cognitive impairment.
	1. Overview of Studies Three original hypotheses were tested during this grant. 1. Sleep perturbation reduces the ability of rats to perform executive functions and will be especially pronounced in space radiation (SR)-exposed rats. 2. Sleep perturbation alters the sensitivity of the brain to develop SR-induced cognitive impairment. 3. SR irradiation alters Delta and Theta wave activity (EEG), and that these SR-induced changes in EEG activity are the underlying cause of the impairment of executive function performance. In 2019, we received supplemental funding to develop the technical capability to simultaneously assess neural activity in multiple brain regions (neural network activity) of rats that are actively engaged in cognitive flexibility tasks, and how sleep fragmentation (SF) impacts neural network cohesiveness. This new concept largely replaced the original third hypothesis after we published data that addressed that hypothesis. In 2020, we received supplemental funding (a NASA Human Research Program (HRP) Investigators' Workshop (IWS) Graduate Fellowship) to establish whether SF exacerbates SR-induced deficits in fine motor skills. In 2021, we received supplemental funding to establish whether SF impacts neural network cohesion (under cognitive loading). 2. Context of Studies: Astronauts on deep space missions will be required to work more autonomously than on previous missions, and thus their ability to deal with unexpected emergencies could be critical to mission success. Decision making is a complex process that is highly dependent upon attentional set-shifting (ATSET) functionality. Unfortunately, multiple ground-based studies have shown that SR exposure significantly impairs ATSET performance in both male and female rats. While roughly a third of the rats have impaired ATSET performance after SR, this may not fully reflect the problem, since astronauts frequently get insufficient restful sleep which often leads to reduced cognitive performance. There is thus a high probability that there will be at least additive cognitive performance decrements over that induced by SR exposure, when astronauts experience sleep perturbations. However, sleep perturbations also increase the tendency to make mistakes and reduces the ability to detect and correct errors. Thus, there is the added possibility that there may be synergistic interactions between SF and SR exposure. Our studies have thus determined how inadequate sleep (SF), either at the time of SR exposure, or 3 months later, alters the incidence, nature, and severity of SR-induced cognitive impairment. The majority of our studies focused on how SR and SF, either alone or in combination, impact ATSET performance, but

Task Progress:	problem solving in humans involves complex situations in which there is a broader, unconstrained set of possible solutions (unconstrained cognitive flexibility). We have thus also established the impact of combined SR and SF on performance in two other advanced tasks: (1) the Unrestrained Cognitive flexibility (UCFlex) assay that requires rats to use creative problem solving to complete the task, and (2) the ARMIT task, an advanced touchscreen-based executive function task that requires rats to use causal reasoning, situational awareness and taxes the ability of the rats to multi-task (i.e., solve a problem in the face of interference). Performance in most neurocognitive tasks requires multiple processes (frequently regulated by several brain regions) to be utilized. Typically, as the complexity of a task increases, so does the number of cognitive processes that are required to successfully complete that task, and thus there is an increased need to efficiently co-ordinate the various cognitive processes. We hypothesized that SR-induced impairment of executive function performance results from dysregulation of the activity of task-positive neural networks, and that the exacerbation of the cognitive impairment following SF stems from further dysregulation of neural network coordination. We have used a highly novel state of the art approach to assess the activity of multiple brain regions in real time as rats perform in our advanced cognitive task. The final aspect of our studies has been to determine the impact of SR and SF, either alone or in combination, has on organized sensorimotor performance during a string pull task. 3. Executive Summary Our studies have provided several novel and significant insights into the impact that combined flights stressors (SR and SF) have on multiple executive functions and organized sensorimotor functions. The data generated in this study have important implications for risk assessment models and highlight the need for detailed studies on the impact of multiple combined space flight stressors on central nervous system (CNS) function. Notable achievements during this grant include: The first studies to demonstrate that SF unveils latent ATSET performance decrements. • This phenomenon occurs in rats exposed to multiple types of SR, including 10 cGy of He, Si, and Galactic Cosmic Ray simulation (GCRsim). It is not seen in rats that have not been exposed to SR. • There is considerable inter-individual variability in the impact that combined SR/SF exposure has on cognitive impairment. • The “unveiled” performance decrements occur in ATSET stages that involve set shifting/reversal functions, which are not observed after exposure to SR or SF alone. • This phenomenon occurs in male and female rats (exposed to 10 cGy GCRsim). • While this phenomenon is observed in both male and female rats exposed to GCRsim, there are sex-dependent differences in the nature of the decrement, and ATSET stage where they are observed. • These data suggest that the true impact of SR-induced cognitive impairment may not be fully evident in normally rested rats, underestimating both the incidence and nature of performance decrements that occur when multiple space flight stressors are present. The first studies to demonstrate that SF unveils latent ARMIT (causal reasoning) performance decrements. • This phenomenon occurs in both male and female rats exposed to 10 cGy of GCRsim. It is not seen in rats that have not been exposed to SR. • The “unveiled” performance decrements suggest that SF leads to a decreased ability to contend with “interference”. The first studies to demonstrate that Local Field Potential (LFP) outputs from the posterior parietal cortex (PPC) and prefrontal cortex (PFC) are collaterally increased when rats are subjected to high cognitive demands during decision making tasks. • This highly novel approach has provided unique insight into how neural network cohesion changes in a dynamic fashion depending upon cognitive demand, with LFP outputs from the PPC and PFC being collaterally increased during de novo decision making tasks. • SF serves to increase the demand for anterograde interference avoidance. The first studies to suggest that SR exposure during periods of ongoing sleep perturbations can negatively alter UCFlex performance. • Our studies have highlighted the severe consequence of inadequate sleep on the frequency and severity of SR-induced impairment of unconstrained (but not constrained) cognitive flexibility. • However, our studies suggest that going to sleep after SR exposure can serve to ameliorate some of the SR-induced cognitive impairments. The first studies to demonstrate that SR and SF independently impact organized sensorimotor function, and that there is merely additivity between these flight stressors (with SF primarily impacting head and postural stability).

Bibliography Type:	Description: (Last Updated: 03/31/2026)
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Articles in Peer-reviewed Journals	Britten RA, Tamgue EN, Arriaga Alvarado P, Fesshaye AS, Sanford LD. "A two-hit model of executive dysfunction: Simulated galactic cosmic radiation primes latent deficits revealed by sleep fragmentation." Life. 2025 Nov 6;15(11):1717. https://doi.org/10.3390/life15111717 ; ; PMID: 41302141 , Nov-2025